PROGRAMMING FUNDAMENTALS

Spring 2024

15th – 21st Feb

Q5

- Ask the user to enter a number as dividend. User also enters another integer value as the divisor. Calculate the quotient and the remainder after dividing the two numbers values. Display dividend, divisor, quotient, and the remainder.

- Hint: declare all numbers as integers

Q6

- Program that accepts height in meters. It then calculates your height in feet. Keep your input and output with decimal format.

- Formula: heightInFeet = heightInMeters * 3.28

Q7

- There are five candidates in your constituency in Feb-2024 elections. Total number of registered voters are 1025. Ask the user to enter the name and the votes for each candidate. Calculate the total votes casted in this constituency. Display the Total votes casted, votes earned by each candidate with his/her name

Q7 - Run Window

```
Enter the name of 1st Candidate : AAA
Enter the votes casted for AAA : 100

Enter the name of 2nd Candidate : BBB
Enter the votes casted for BBB : 200

Enter the name of 3rd Candidate : CCC
Enter the votes casted for CCC : 300

Enter the name of 4th Candidate : DDD
Enter the votes casted for DDD : 400

Enter the name of 5th Candidate : EEE
Enter the votes casted for EEE : 500

Votes taken by AAA = 100
Votes taken by BBB = 200
Votes taken by CCC = 300
Votes taken by DDD = 400
Votes taken by EEE = 500
Total Votes Casted in this Constituency = 1500
-----------------------------------
```

Q8

- Declare three string variables and store the following quotes:

1. Computers have lots of memory but no imagination: BillGates
2. Isaac = "I do not fear computers. I fear the lack of them : Isaac Asimov
3. Man is still the most extraordinary computer of all : John F. Kennedy

Display the three quotes on separate lines using the string variables

Q8 Hints

```
cout<<Bill<<endl;
cout<<Isaac<<endl;
cout<<Kennedy<<endl;
```

```
Computers have lots of memory but no imagination
I do not fear computers. I fear the lack of them
Man is still the most extraordinary computer of all
```

Selection Structure

- IF statement
- Syntax

```
if (expression)
     statement1
else
     statement2
```

Q1

- Program to calculate the wages of an employee based on the number of hours worked. If the employee worked for 40 or less number of hours in a week, wages will be calculated as:

wages = hours * HourlyRate

- If the employee works for more than 40 hours, then the formula will be as follows:

wages = 40.0 * HourlyRate + 1.5 * HourlyRate * (hours – 40.0)

- Use double as data type for your variables before accepting inputs.

```
if (hours > 40.0)
    wages = 40.0 * rate +
            1.5 * rate * (hours - 40.0);
else
    wages = hours * rate;
```

Q2

- Ask the user to enter the temperature. If the temperature is greater than 25^0C and less than 33^0C then display a message “Good day for Tennis”. Otherwise display message “You must stay inside”.

```
Enter the temperature = 30
Good Day for Tennis
```

Q3

- Ask the user to enter word count of the essay written by him/her. If the word count is greater than 2500, display the message “You must review your essay” otherwise, display the message “Your essay is accepted for the competition”.

```
Enter the Word Count of your essay = 2350
Your essay is accepted for the competition..
```

```
Enter the Word Count of your essay = 3120
You must review your essay
```

Q4

- Write an if statement that performs the following logic: if the sales entered by the user are greater or equal to than Rs. 50,000, then assign 0.25 to the CommRate variable, and assign 250 to the bonus and display Sales, Commision (Sales * CommRate), and Bonus.
- Otherwise simply display Sales and a message that you did not earn any additional commission and bonus with a single cout statement.

Q4 Run window

```
Enter the Sales Amount = 45000
Your Sales Amount = 45000
You did not earn any Additional Commission
You got NO Bonus
```

```
Enter the Sales Amount = 52000
Your Sales Amount = 52000
Commission you Earned = 13000
Your Bonus = 250
```

Q5

- Write a program that prompts the user to input a number (an integer). The program should then output a message saying whether the number is a 2-digit positive number or a 3-digit positive number or any other number.

Q5 – Run Window

```
Enter a number 68
Its a two digit positive number
```

```
Enter a number 651
Its a three digit positive number
```

Q6 – Practice elseif

- Ask the user to enter an integer from 1 and 12 (inclusive). Display the name of the month based on the number value entered by the user. If the user enters 2 then display “February” and so on.
- Complete the code below.

```
if (month == 1)
    cout << "January" << endl;
else if (month == 2)
    cout << "February" << endl;
```

Q7

- Write a program that prompts the user to input three numbers. The program should then output the numbers in ascending order.

Q8

- Write a program that prompts the user to input an integer between **0** and **15**. If the number is less than or equal to **9**, the program should output the number; otherwise, it should output **A** for **10**, **B** for **11**, **C** for **12**, D for 13, E for 14 and F for 15.
- Otherwise display “Invalid input”

Q9

- In a right triangle, the square of the length of one side is equal to the sum of the squares of the lengths of the other two sides.
- Write a program that prompts the user to enter the lengths of three sides of a triangle and then outputs a message indicating whether the triangle is a right triangle.

Q10

- Ask the user to enter two integers. The first integer is the dividend
- The second integer is the divisor
- If the divisor is zero, display the message that “division by zero is not possible. Run the program again”
- Otherwise, carry out the division and dislay the answer in decimal number format

Book Reference

- D.S. Malik (ch – 4)
- Tony Gaddis (ch - 4)